



SARDAR PATEL UNIVERSITY
B. C. A EXAMINATION, VI SEM

Date : 08th Apr 2022, Friday

Time : 12:00pm To 02:00pm

Session : NOON

Sub: System Software

Course No : US06FBCA01

Total marks : 70

Q – 1 Multiple Choice Question

[10]

- i) _____ part build the desired target program by using intermediate representation of two part.
 (a) Analysis (b) Synthesis (c) Execution (d) Intermediate Code
- ii) _____ part of compilation creates an intermediate representation of the source program.
 (a) Synthesis (b) Analysis (c) Intermediate code (d) Execution
- iii) _____ associates meaning with valid statements of the language.
 (a) Specification Rules (b) Lexical Rules
 (c) Syntax Rules (d) Semantic Rules
- iv) _____ data structure is used in synthesis phase.
 (a) DFD (b) Symbol Table (c) Name table (d) All of the above
- v) The _____ performs memory bindings to symbolic names.
 (a) Compiler (b) Translator (c) Assembler (d) Analyser
- vi) In a simple assembly language, the First operand is always _____.
 (a) Register (b) Data declaration (c) Memory word (d) Mnemonic
- vii) Address of the origin assumed by the translator is known as _____.
 (a) Linked Origin (b) Translated Origin (c) Load Origin (d) Origin
- viii) The _____ maintains an internal form which is used to perform the edit operation.
 (a) Linker (b) Loader (c) Editor (d) None of these
- ix) _____ helps the user in selecting test data for his program..
 (a) Test Data Generator (b) Language Processor
 (c) Software Tools (d) Program Environment
- x) _____ helps in obtaining information for localization of errors..
 (a) Profile monitors (b) Debug Monitors (c) Chosen text (d) Editors

Q – 2 Fill in the Blanks and True – False

[08]

- i) _____ part of compilation breaks up the source program into essential pieces.
- ii) The gap between PL domain and Execution domain is called execution gap. (True / False)
- iii) _____ is a data structure to implement memory allocation.
- iv) Symbol table has Name and Address primary fields. (True / False)
- v) Word processors are also called _____ editors.
- vi) Linking is the process of binding an external reference to the correct link time address. (True / False)
- vii) _____ is responsible for managing the user's screen and accepting data and presenting results.
- viii) Data mode accepts user commands specifying the editing function to be performed. (True / False)

Q – 3 Short Answer attempt any ten (Each carry 2 marks)

[20]

- i) What do you mean by compiler
- ii) Define : Application Domain and Execution Domain
- iii) Draw a diagram of a language processing system
- iv) Give instruction opcode and function of following assembly mnemonic.
1) BC 2) DIV 3) MOVER 4) PRINT
- v) What is machine instruction format?
- vi) Explain operand specification <Operand spec> with its syntax.
- vii) List out different types of editors
- viii) Explain address sensitive program.
- ix) Define ENTRY and EXTRN statement of public definition.
- x) Define program generator
- xi) Define Program Environment
- xii) Explain Debug Monitor

Q – 4 Attempt any four (Each carry 8 marks)

[32]

- i) Explain compiler phases in detail.
- ii) What is language processing? Explain Application, PL & Execution domain.
- iii) Explain Analysis and Synthesis phase of Assembly language.
- iv) Write note on Assembly language with it's types in detail.
- v) Explain all the editors in detail
- vi) Explain Translated, Linked and load time address in detail.
- vii) Define software tools. Discuss software tools for programming development.
- viii) Explain structure of user interface and it's management system.

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